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**EE-802 (GS)**  
**B.E. VIII Semester** Examination, June 2020  
**Grading System (GS)**  
**Power System Protection**  
*Time : Three Hours*  
*Maximum Marks :70*

**Note :** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. What is the principle of differential relays? What are their limitation? Explain their characteristics.
2. Derive the expression of the symmetrical components of fault current of a power system for L-L fault through impedance. A 30 MVA, 11 kV generator has  $Z_1=Z_2=j0.2$  pu,  $Z_0=j0.05$  pu. A line to ground fault occurs on the generator terminals. Find the fault current and line to line voltages during fault conditions.
3. Explain the following terms as applied to protective relays:
  - i) Plug Setting Multiplier (PSM)
  - ii) Resetting time
  - iii) Burden
  - iv) Back up relay
  - v) Over reach
4. Explain the construction, working principle and characteristics of reactance relay type distance relay.
5. Explain the following protection schemes of alternator.
  - i) Stator inter turn protection
  - ii) Merz price protection.
6. A 3-phase transformer of 220/11000V line volts is connected in star delta. The protective transformer on 220V side have a current ratio of 600/5. What should be the CT ratio on 11000V side.
7. Briefly describe a microprocessor based scheme for automatic load shedding and restoration.

OR

What is meant by directional over current relay? Describe its construction and principle of operating.
8. Write short notes on any two of the following:
  - a) Static relay functional circuits
  - b) Rating selection of circuit breaker
  - c) Transmission line protection
  - d) Sequence networks for unsymmetrical faults

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